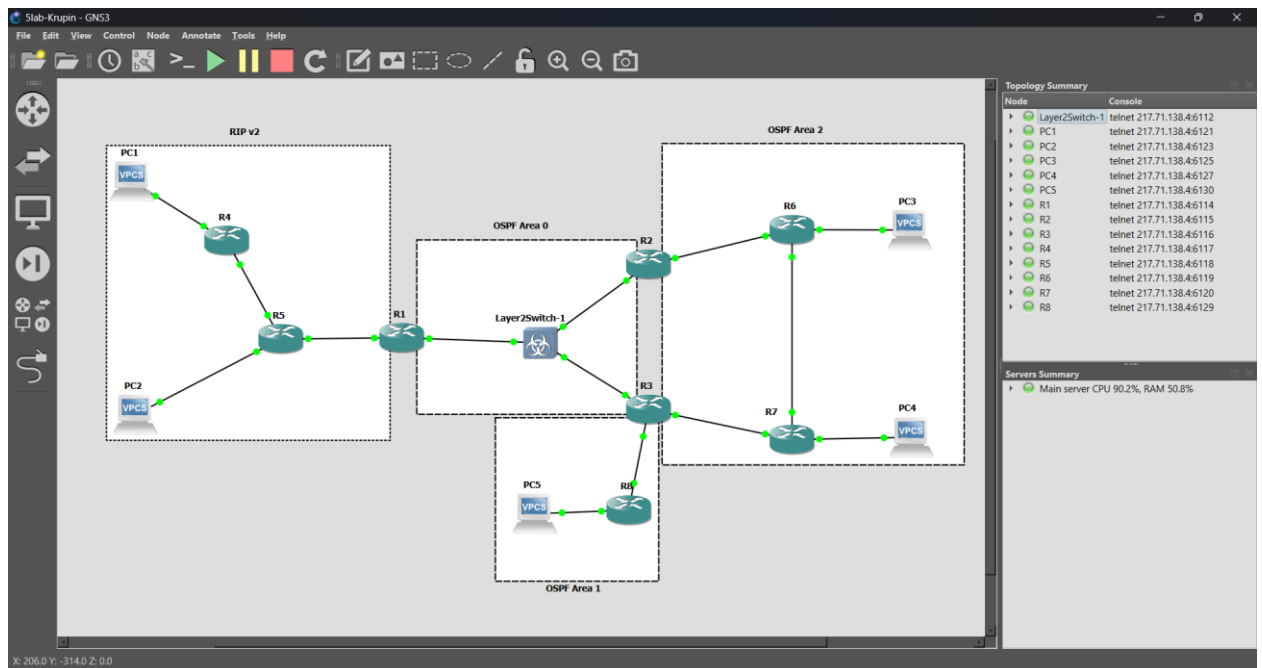




Планирование адресов

Перед настройкой протоколов динамической маршрутизации было выполнено планирование адресного пространства сети. В соответствии с условием задания каждое соединение между двумя маршрутизаторами является отдельной сетью. Для таких соединений используются подсети /30 (255.255.255.252), поскольку в них требуется только два IP-адреса — по одному для каждого маршрутизатора.

Для локальных сетей, к которым подключены персональные компьютеры, используются подсети /24 (255.255.255.0). Это позволяет при необходимости подключать дополнительные устройства без изменения схемы адресации. В области OSPF Area 0 маршрутизаторы R1, R2 и R3 подключены через коммутатор второго уровня и находятся в одной общей сети, поэтому для них используется одна подсеть 10.0.10.0/24.

LAN1 (PC1 – R4)
192.168.10.0/24

LAN2 (PC2 – R5)
192.168.20.0/24

LAN3 (PC3 – R6)
192.168.30.0/24

LAN4 (PC4 – R7)
192.168.40.0/24

LAN5 (PC5 – R8)
192.168.50.0/24

Сети между маршрутизаторами

R4 – R5
10.0.1.0/30

R5 – R1
10.0.2.0/30

R2 – R6
10.0.3.0/30

R6 – R7
10.0.4.0/30

R7 – R3
10.0.5.0/30

R3 – R8
10.0.6.0/30

Общая сеть OSPF Area 0

R1, R2, R3
10.0.10.0/24

В консоль R1 вводим

enable

configure terminal

show ip interface brief

interface FastEthernet0/0

no shutdown

exit

interface FastEthernet1/0

no shutdown

exit

exit

show ip interface brief

```
R1
S - Switch, H - Host, I - IGMP, r - Repeater

Device ID      Local Intrfce    Holdtme    Capability  Platform  Port ID
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Prot
FastEthernet0/0 unassigned      YES unset  up          up
FastEthernet1/0 unassigned      YES unset  administratively down down
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet1/0
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar  1 05:28:10.974: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar  1 05:28:11.974: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config-if)#exit
R1(config)#exit
R1#
*Mar  1 05:28:16.770: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Prot
FastEthernet0/0 unassigned      YES unset  up          up
FastEthernet1/0 unassigned      YES unset  up          up
R1#
R1#
```

В консоль R5 вводим

enable

show ip interface brief

configure terminal

interface FastEthernet 0/0

no shutdown

exit

interface FastEthernet 1/0

no shutdown

exit

interface FastEthernet 2/0

no shutdown

exit

exit

show cdp neighbors

```
R5
R5(config)#exit
R5#
*Mar  1 05:26:49.526: %SYS-5-CONFIG_I: Configured from console by console
R5#show cdp neighbors
^
% Invalid input detected at '^' marker.

R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#show cdp neighbors
^
% Invalid input detected at '^' marker.

R5(config)#exit
R5#
*Mar  1 05:27:45.986: %SYS-5-CONFIG_I: Configured from console by console
R5#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
R1                 Fas 2/0         147        R S I       3640       Fas 0/0
R5#
R5#
```

Настраиваем R5

configure terminal

interface FastEthernet2/0

ip address 10.0.2.1 255.255.255.252

no shutdown

exit

exit

Проверяем R5

show ip interface brief

```
R5#
R5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#
R5(config)#interface FastEthernet2/0
R5(config-if)#
R5(config-if)#ip address 10.0.2.1 255.255.255.252
R5(config-if)#
R5(config-if)#no shutdown
R5(config-if)#
R5(config-if)#exit
R5(config)#exit
R5#
*Mar  1 05:45:17.762: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip interface brief
Interface                               IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0                         unassigned      YES unset    up
FastEthernet1/0                         unassigned      YES unset    up
FastEthernet2/0                         10.0.2.1        YES manual   up
R5#
```

R1

enable

configure terminal

interface FastEthernet0/0

ip address 10.0.2.2 255.255.255.252

no shutdown

exit

exit

Проверяем R1

show ip interface brief

configure terminal

interface FastEthernet1/0

ip address 10.0.10.1 255.255.255.0

no shutdown

exit

exit

Проверяем R1

show ip interface brief

```
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 10.0.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
*Mar  1 05:59:10.530: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface                               IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0                        10.0.2.2        YES manual  up          up
FastEthernet1/0                        10.0.10.1       YES manual  up          up

R1#
R1#
R1#
R1#
R1#
R1#
```

В консоли R1 выполняем

ping 10.0.2.1

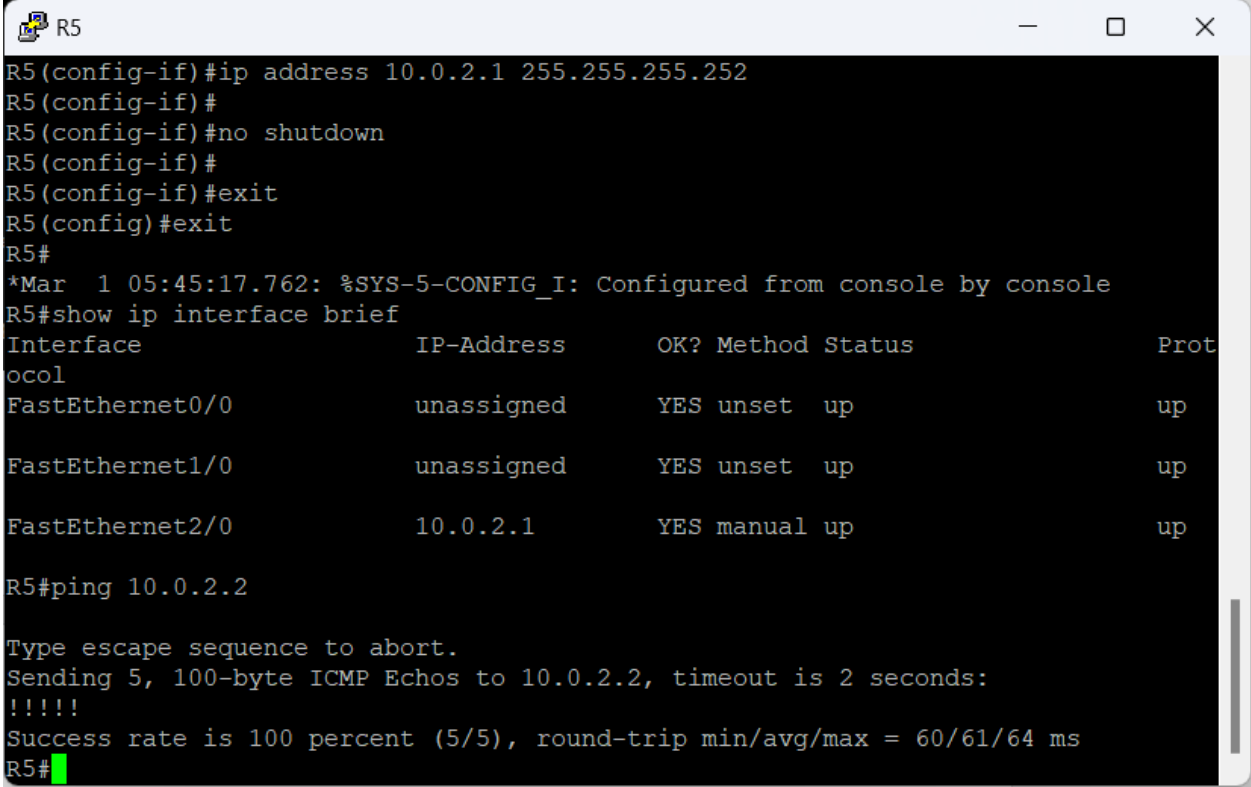
```
R1(config-if)#
R1(config-if)#exit
R1(config)#exit
R1#
*Mar  1 05:59:10.530: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip interface brief
Interface                               IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0                        10.0.2.2        YES manual  up          up
FastEthernet1/0                        10.0.10.1       YES manual  up          up

R1#
R1#
R1#
R1#
R1#
R1#ping 10.0.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 60/65/76 ms
R1#
```

В консоли R5

ping 10.0.2.2



```
R5
R5(config-if)#ip address 10.0.2.1 255.255.255.252
R5(config-if)#
R5(config-if)#no shutdown
R5(config-if)#
R5(config-if)#exit
R5(config)#exit
R5#
*Mar  1 05:45:17.762: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip interface brief
Interface                IP-Address      OK? Method Status  Prot
ocol
FastEthernet0/0          unassigned      YES unset  up      up
FastEthernet1/0          unassigned      YES unset  up      up
FastEthernet2/0          10.0.2.1        YES manual  up      up

R5#ping 10.0.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/61/64 ms
R5#
```

Настраиваем R2

enable

show ip interface brief

configure terminal

interface FastEthernet0/0

no shutdown

exit

interface FastEthernet1/0

no shutdown

exit

exit

show cdp neighbors

Переходим на R6

enable

show ip interface brief

configure terminal

```
interface FastEthernet0/0
no shutdown
exit
```

```
interface FastEthernet1/0
no shutdown
exit
```

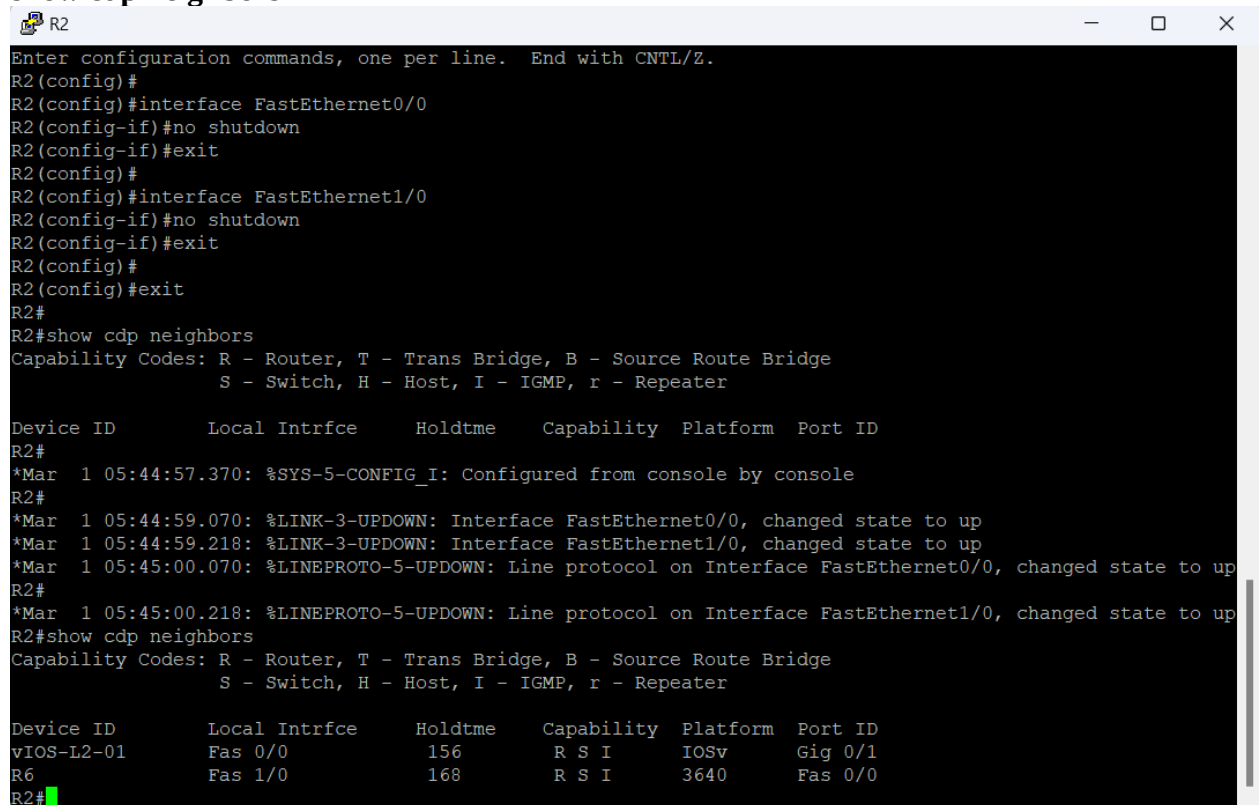
```
interface FastEthernet2/0
no shutdown
exit
```

```
exit
```

```
show cdp neighbors
```

Переходим на R2

```
show cdp neighbors
```



```
R2
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#
R2(config)#interface FastEthernet0/0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#interface FastEthernet1/0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
R2(config)#exit
R2#
R2#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID         Local Intrfce   Holdtme    Capability Platform Port ID
R2#
*Mar  1 05:44:57.370: %SYS-5-CONFIG_I: Configured from console by console
R2#
*Mar  1 05:44:59.070: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 05:44:59.218: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar  1 05:45:00.070: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2#
*Mar  1 05:45:00.218: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R2#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID         Local Intrfce   Holdtme    Capability Platform Port ID
vIOS-L2-01        Fas 0/0         156        R S I     IOSv     Gig 0/1
R6                 Fas 1/0         168        R S I     3640     Fas 0/0
R2#
```

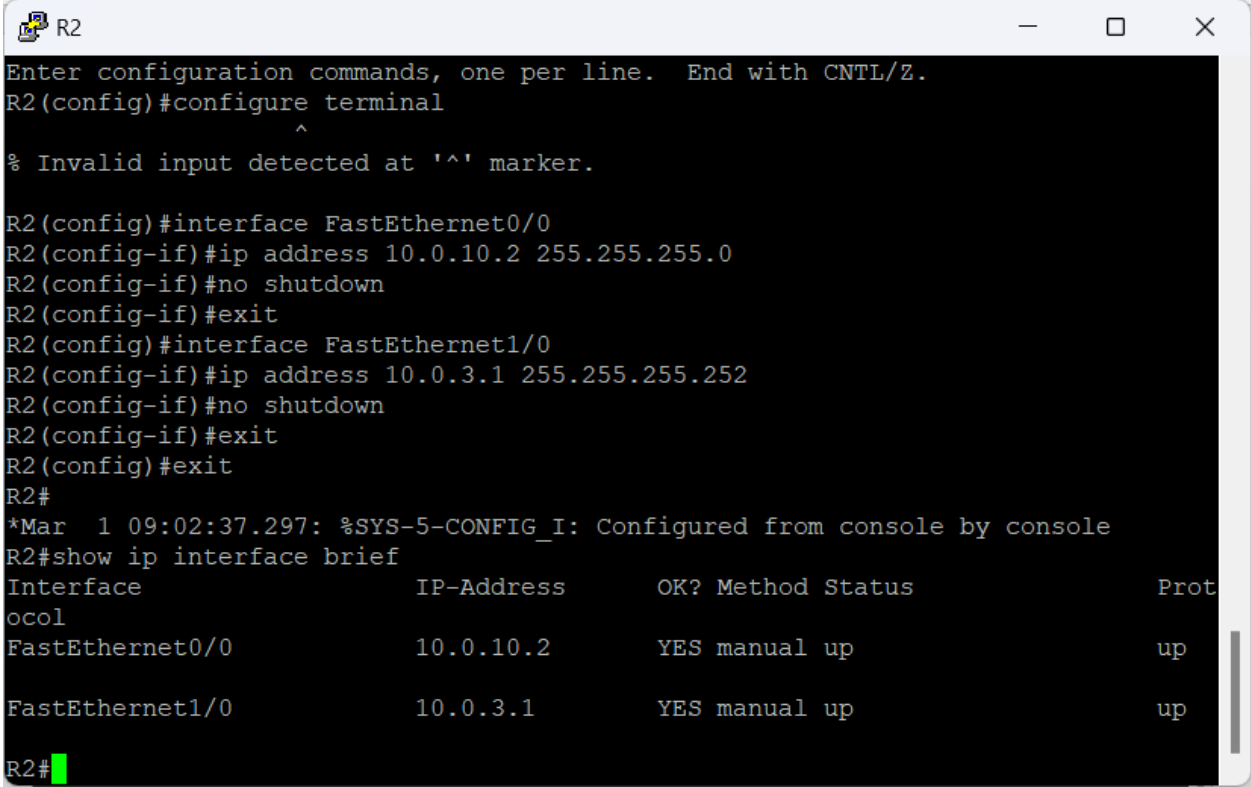
R2

configure terminal

```
interface FastEthernet0/0
ip address 10.0.10.2 255.255.255.0
no shutdown
exit
interface FastEthernet1/0
ip address 10.0.3.1 255.255.255.252
no shutdown
```


exit
exit

show ip interface brief



```
R2
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#configure terminal
^
% Invalid input detected at '^' marker.

R2(config)#interface FastEthernet0/0
R2(config-if)#ip address 10.0.10.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#interface FastEthernet1/0
R2(config-if)#ip address 10.0.3.1 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#exit
R2#
*Mar  1 09:02:37.297: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          10.0.10.2        YES manual  up          up
FastEthernet1/0          10.0.3.1         YES manual  up          up
R2#
```

На R6 вводим

enable
configure terminal
interface FastEthernet0/0
ip address 10.0.3.2 255.255.255.252
no shutdown
exit
exit

show ip interface brief

Ha R2
ping 10.0.3.2
Ha R6
ping 10.0.3.1


```
R6#
R6#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID      Local Intrfce   Holdtme    Capability  Platform  Port ID
R2              Fas 0/0         141        R S I       3640      Fas 1/0
R7              Fas 2/0         161        R S I       3640      Fas 2/0
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
```

Ha R6

enable

configure terminal

interface FastEthernet2/0

ip address 10.0.4.1 255.255.255.252

no shutdown

exit

exit

show ip interface brief

```
R6#
*Mar  1 09:02:57.801: %SYS-5-CONFIG_I: Configured from console by console
R6#enable
R6#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#interface FastEthernet2/0
R6(config-if)#ip address 10.0.4.1 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#exit
R6#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          10.0.3.2        YES manual up          up
FastEthernet1/0          unassigned      YES manual up          up
FastEthernet2/0          10.0.4.1        YES manual up          up

R6#
*Mar  1 09:03:14.221: %SYS-5-CONFIG_I: Configured from console by console
R6#
R6#
```

Ha R7

enable

configure terminal

interface FastEthernet2/0

ip address 10.0.4.2 255.255.255.252

no shutdown

exit

exit

show ip interface brief

```
R7
R7#enable
R7#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R7(config)#
R7(config)#interface FastEthernet2/0
R7(config-if)#ip address 10.0.4.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#exit
R7#
R7#show ip interface brief
Interface IP-Address OK? Method Status Prot
ocol
FastEthernet0/0 unassigned YES manual up
FastEthernet1/0 unassigned YES unset up
FastEthernet2/0 10.0.4.2 YES manual up

R7#
*Mar 1 08:47:49.493: %SYS-5-CONFIG_I: Configured from console by console
R7#
R7#
R7#
```

На R7 вводим
ping 10.0.4.1

На R6 вводим
ping 10.0.4.2

```
R6
R6(config)#interface FastEthernet2/0
R6(config-if)#ip address 10.0.4.1 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#exit
R6#show ip interface brief
Interface IP-Address OK? Method Status Prot
ocol
FastEthernet0/0 10.0.3.2 YES manual up
FastEthernet1/0 unassigned YES manual up
FastEthernet2/0 10.0.4.1 YES manual up

R6#
*Mar 1 09:03:14.221: %SYS-5-CONFIG_I: Configured from console by console
R6#
R6#ping 10.0.4.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.4.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 56/60/64 ms
R6#

R7
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#exit
R7#
R7#show ip interface brief
Interface IP-Address OK? Method Status Prot
ocol
FastEthernet0/0 unassigned YES manual up
FastEthernet1/0 unassigned YES unset up
FastEthernet2/0 10.0.4.2 YES manual up

R7#
*Mar 1 08:47:49.493: %SYS-5-CONFIG_I: Configured from console by console
R7#
R7#
R7#ping 10.0.4.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.4.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/67/92 ms
R7#
```

На R3

show ip interface brief
enable
configure terminal

interface FastEthernet0/0
no shutdown
exit

interface FastEthernet1/0
no shutdown

exit

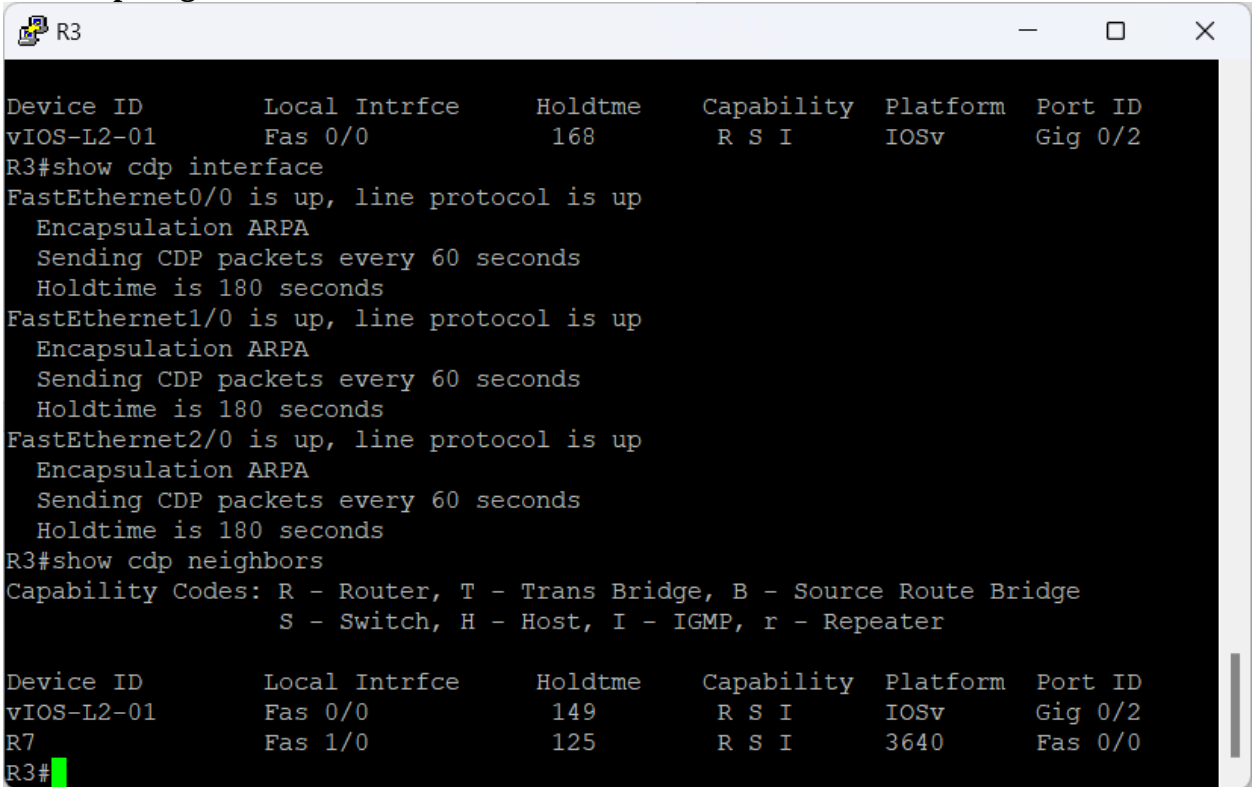
interface FastEthernet2/0

no shutdown

exit

exit

show cdp neighbors



```
R3
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
vIOS-L2-01     Fas 0/0        168      R S I       IOSv       Gig 0/2
R3#show cdp interface
FastEthernet0/0 is up, line protocol is up
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
FastEthernet1/0 is up, line protocol is up
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
FastEthernet2/0 is up, line protocol is up
  Encapsulation ARPA
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
R3#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
vIOS-L2-01     Fas 0/0        149      R S I       IOSv       Gig 0/2
R7             Fas 1/0        125      R S I       3640       Fas 0/0
R3#
```

Ha R8

enable

configure terminal

interface FastEthernet0/0

no shutdown

exit

interface FastEthernet1/0

no shutdown

exit

exit

show cdp neighbors

```
R8
R8(config)#interface FastEthernet1/0
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
R8(config)#exit
*Mar 1 07:18:43.374: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 07:18:43.522: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 07:18:44.374: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R8(config)#exit
*Mar 1 07:18:44.522: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R8(config)#exit
R8#configure terminal
*Mar 1 07:19:00.418: %SYS-5-CONFIG_I: Configured from console by console
R8#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID         Local Intrfce   Holdtme    Capability  Platform  Port ID
R3                 Fas 0/0         179        R S I       3640      Fas 2/0
R8#
```

Ha R3

enable
configure terminal

interface FastEthernet2/0
ip address 10.0.6.1 255.255.255.252
no shutdown
exit

exit

Ha R8

enable
configure terminal

interface FastEthernet0/0
ip address 10.0.6.2 255.255.255.252
no shutdown
exit

exit

show ip interface brief

```
R8

S - Switch, H - Host, I - IGMP, r - Repeater

Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
R3             Fas 0/0       138      R S I       3640      Fas 2/0

R8#enable
R8#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#
R8(config)#interface FastEthernet0/0
R8(config-if)#ip address 10.0.6.2 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
R8(config)#exit
R8#
*Mar  1 07:27:37.574: %SYS-5-CONFIG_I: Configured from console by console
R8#show ip interface brief
Interface      IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0  10.0.6.2        YES manual up          up
FastEthernet1/0  unassigned      YES unset  up          up

R8#
R8#
```

Ha R3
ping 10.0.6.2

Ha R8
ping 10.0.6.1

```
R3
R3#0
R3#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater

Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
vIOS-L2-01     Fas 0/0       162      R S I       IOSv      Gig 0/2
R7             Fas 1/0       121      R S I       3640      Fas 0/0
R8             Fas 2/0       158      R S I       3640      Fas 0/0

R3#show ip interface brief
Interface      IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0  unassigned      YES unset  up          up
FastEthernet1/0  unassigned      YES unset  up          up
FastEthernet2/0  10.0.6.1        YES manual up          up

R3#ping 10.0.6.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.6.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 36/54/60 ms
R3#

R8
Enter configuration commands, one per line.  End with CNTL/Z.
R8(config)#
R8(config)#interface FastEthernet0/0
R8(config-if)#ip address 10.0.6.2 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
R8(config)#exit
R8#
*Mar  1 07:27:37.574: %SYS-5-CONFIG_I: Configured from console by console
R8#show ip interface brief
Interface      IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0  10.0.6.2        YES manual up          up
FastEthernet1/0  unassigned      YES unset  up          up

R8#
R8#ping 10.0.6.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.6.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/54/60 ms
R8#
```

Ha R3

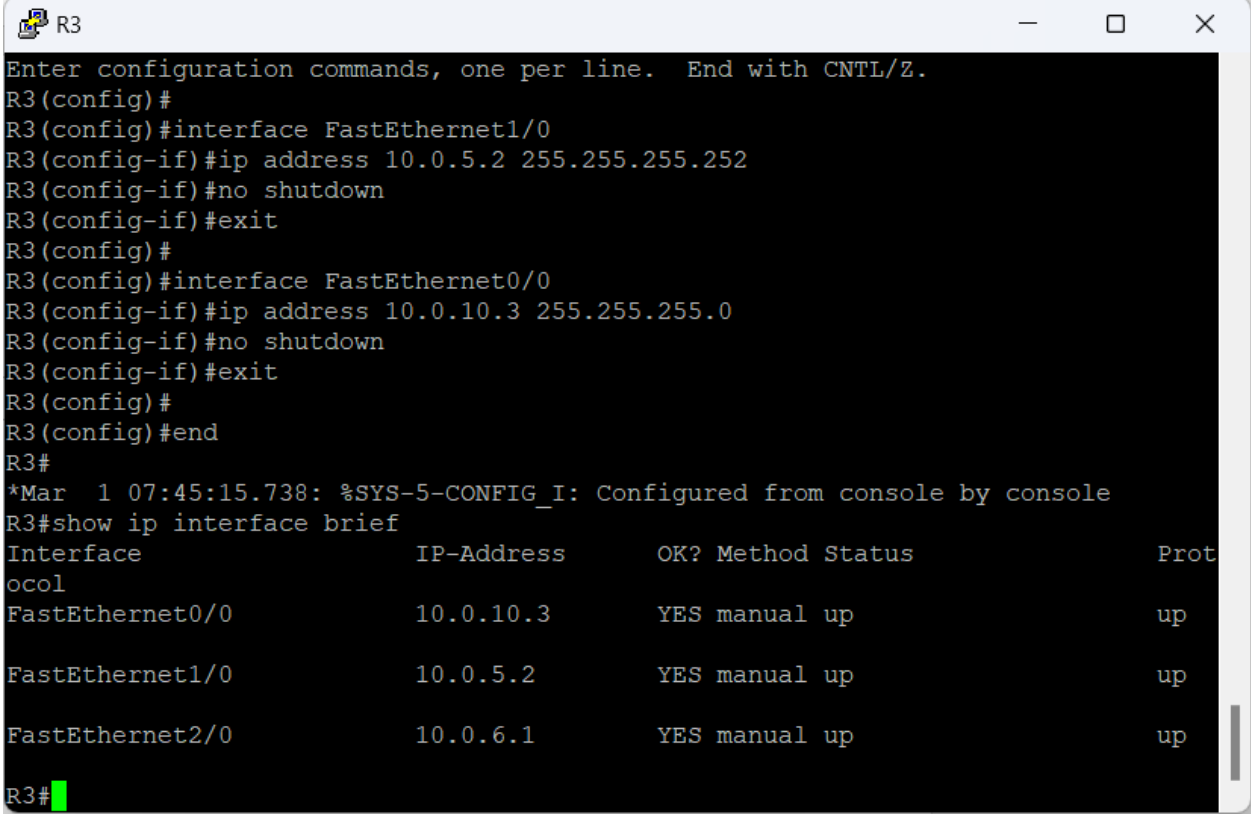
enable
configure terminal

interface FastEthernet1/0
ip address 10.0.5.2 255.255.255.252
no shutdown
exit

interface FastEthernet0/0


```
ip address 10.0.10.3 255.255.255.0
no shutdown
exit
```

```
end
show ip interface brief
```



```
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#interface FastEthernet1/0
R3(config-if)#ip address 10.0.5.2 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#interface FastEthernet0/0
R3(config-if)#ip address 10.0.10.3 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
R3(config)#end
R3#
*Mar  1 07:45:15.738: %SYS-5-CONFIG_I: Configured from console by console
R3#show ip interface brief
Interface                        IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0                  10.0.10.3       YES manual  up          up
FastEthernet1/0                  10.0.5.2        YES manual  up          up
FastEthernet2/0                  10.0.6.1        YES manual  up          up
R3#
```

Ha R7

```
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 10.0.5.1 255.255.255.252
no shutdown
exit
```

```
end
```

```
R7
R7(config)#
R7(config)#interface FastEthernet0/0
R7(config-if)#ip address 10.0.5.1 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
R7(config)#end
R7#
*Mar  1 00:02:34.571: %SYS-5-CONFIG_I: Configured from console by console
R7#
*Mar  1 00:02:36.411: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:02:37.411: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R7#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.5.1        YES manual  up          up
FastEthernet1/0          unassigned      YES unset   administratively down down
FastEthernet2/0          10.0.4.2        YES manual  up          up
R7#
```

Ha R7
ping 10.0.5.2

```
R7
R7(config)#end
R7#
*Mar  1 00:02:34.571: %SYS-5-CONFIG_I: Configured from console by console
R7#
*Mar  1 00:02:36.411: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:02:37.411: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R7#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          10.0.5.1        YES manual  up          up
FastEthernet1/0          unassigned      YES unset   administratively down down
FastEthernet2/0          10.0.4.2        YES manual  up          up
R7#ping 10.0.5.2

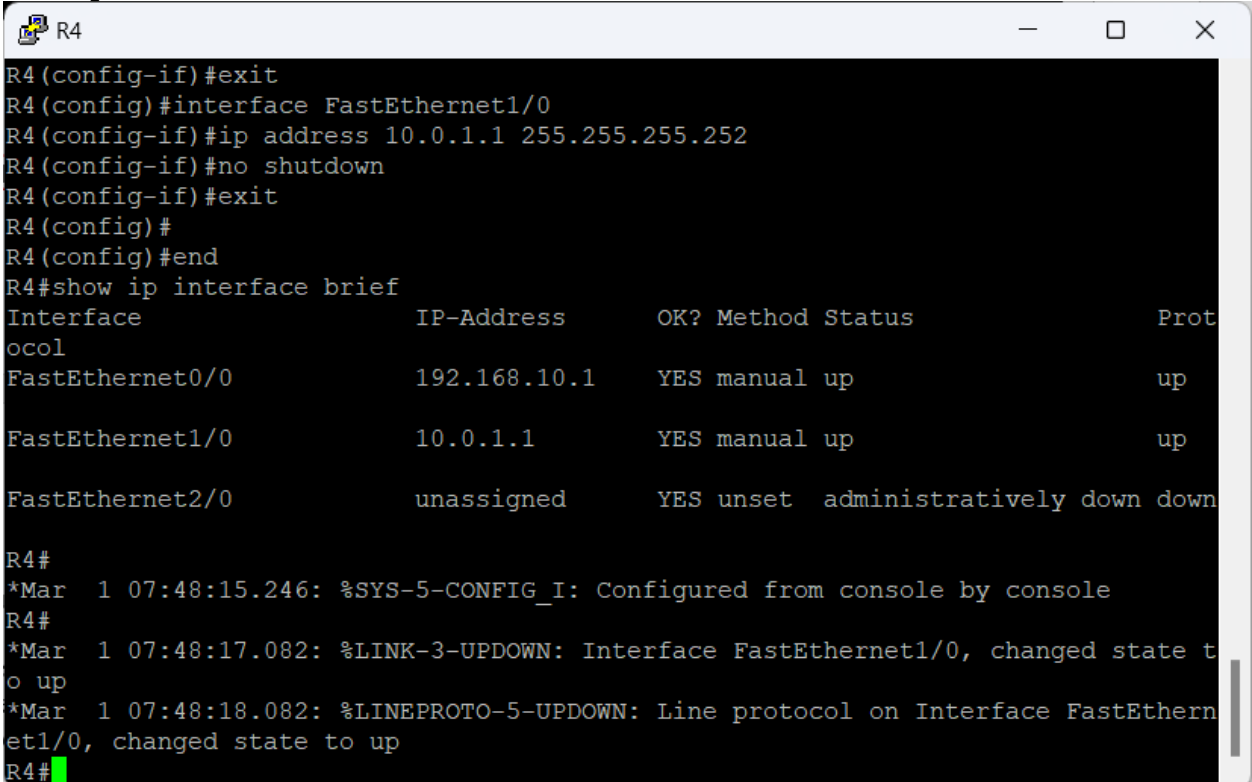
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.5.2, timeout is 2 seconds:
.....
Success rate is 80 percent (4/5), round-trip min/avg/max = 44/55/60 ms
R7#
```

Ha R4
enable
show ip interface brief

show cdp neighbors
enable
configure terminal

interface FastEthernet0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
interface FastEthernet1/0
ip address 10.0.1.1 255.255.255.252
no shutdown
exit

end
show ip interface brief



```
R4
R4(config-if)#exit
R4(config)#interface FastEthernet1/0
R4(config-if)#ip address 10.0.1.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
R4(config)#end
R4#show ip interface brief
Interface                IP-Address      OK? Method Status      Prot
ocol
FastEthernet0/0          192.168.10.1    YES manual  up          up
FastEthernet1/0          10.0.1.1        YES manual  up          up
FastEthernet2/0          unassigned      YES unset   administratively down down

R4#
*Mar  1 07:48:15.246: %SYS-5-CONFIG_I: Configured from console by console
R4#
*Mar  1 07:48:17.082: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar  1 07:48:18.082: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R4#
```

Ha R5
enable
configure terminal

interface FastEthernet1/0
ip address 10.0.1.2 255.255.255.252
no shutdown
exit

end

Ha R4
ping 10.0.1.1

```
R4
R4(config)#end
R4#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.168.10.1 YES manual up up
FastEthernet1/0 10.0.1.1 YES manual up up
FastEthernet2/0 unassigned YES unset administratively down down
R4#
*Mar 1 07:48:15.246: %SYS-5-CONFIG_I: Configured from console by console
R4#
*Mar 1 07:48:17.082: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar 1 07:48:18.082: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R4#ping 10.0.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
R4#
```

PC1

ip 192.168.10.2/24 192.168.10.1

show ip

PC2

ip 192.168.20.2/24 192.168.20.1

show ip

PC3

ip 192.168.30.2/24 192.168.30.1

show ip

PC4

ip 192.168.40.2/24 192.168.40.1

show ip

PC5

ip 192.168.50.2/24 192.168.50.1

show ip

R5

enable

configure terminal

interface FastEthernet0/0

```
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
exit
```

```
R6
enable
configure terminal
```

```
interface FastEthernet1/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
exit
```

```
R7
enable
configure terminal
```

```
interface FastEthernet1/0
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
exit
```

```
R8
enable
configure terminal
```

```
interface FastEthernet1/0
ip address 192.168.50.1 255.255.255.0
no shutdown
exit
exit
```

```
RIP v2
```

```
R4
```

```
enable
configure terminal
router rip
version 2
no auto-summary
network 192.168.10.0
network 10.0.1.0
exit
end
```

```
show ip protocols
```

R5

```
enable  
configure terminal  
router rip  
version 2  
no auto-summary  
network 10.0.0.0  
network 192.168.20.0  
exit  
end
```

show ip protocols

R4

```
show ip interface brief  
show ip protocols  
show ip route
```

R5

```
show ip interface brief  
show ip protocols  
show ip route
```

Ha R1

```
configure terminal  
router rip  
version 2  
no auto-summary  
network 10.0.0.0  
exit  
end
```

show ip protocols

show ip route

Настройка OSPF

R1

```
enable  
configure terminal  
router ospf 1  
network 10.0.10.0 0.0.0.255 area 0  
exit  
end
```

show ip protocols

show ip ospf neighbor

```
R1
Default version control: send version 2, receive version 2
  Interface      Send  Recv  Triggered RIP  Key-chain
FastEthernet0/0    2    2
FastEthernet1/0    2    2
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
Routing Information Sources:
  Gateway         Distance      Last Update
  10.0.2.1         120           00:00:03
Distance: (default is 120)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 10.0.10.1

R1#show ip ospf neighbor

R1#show ip ospf neighbor

R1#
R1#
```

На R2 вводим

```
enable
configure terminal
router ospf 1
network 10.0.10.0 0.0.0.255 area 0
exit
end
```

```
show ip ospf neighbor
```

```
R2
FastEthernet0/0      10.0.10.2      YES manual up      up
FastEthernet1/0      10.0.3.1      YES manual up      up
R2#ping 10.0.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.3.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 36/48/60 ms
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router ospf 1
R2(config-router)#network 10.0.10.0 0.0.0.255 area 0
R2(config-router)#exit
R2(config)#end
R2#
*Mar  1 11:46:21.081: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip ospf neighbor
*Mar  1 11:46:35.517: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.10.1 on FastEthernet0/
0 from LOADING to FULL, Loading Done
R2#show ip ospf neighbor
R2#
```

На R3 вводим:

```
enable
configure terminal
router ospf 1
network 10.0.10.0 0.0.0.255 area 0
exit
end

show ip ospf neighbor
```



```
R3
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 10.0.10.0 0.0.0.255 area 0
R3(config-router)#exit
R3(config)#end
R3#
*Mar  1 09:45:23.945: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Mar  1 09:45:27.173: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.10.2 on FastEthernet0/
0 from LOADING to FULL, Loading Done
R3#
*Mar  1 09:45:32.505: %OSPF-5-ADJCHG: Process 1, Nbr 10.0.10.1 on FastEthernet0/
0 from LOADING to FULL, Loading Done
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.0.10.1        1     FULL/DR         00:00:39    10.0.10.1      FastEthernet0/
0
10.0.10.2        1     FULL/BDR        00:00:37    10.0.10.2      FastEthernet0/
0
R3#
```

R2

```
configure terminal
router ospf 1
network 10.0.3.0 0.0.0.3 area 2
exit
end
```

На R6 вводим

```
enable
configure terminal
router ospf 1
network 10.0.3.0 0.0.0.3 area 2
exit
end
```

```
show ip ospf neighbor
```

```
R6
R6#enable
R6#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R6(config)#router ospf 1
R6(config-router)#network 10.0.3.0 0.0.0.3 area 2
R6(config-router)#exit
R6(config)#end
R6#
*Mar  1 11:55:31.085: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.0.10.2         1    FULL/DR         00:00:33    10.0.3.1       FastEthernet0/
0
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
R6#
```

R7

```
enable
configure terminal
router ospf 1
network 10.0.4.0 0.0.0.3 area 2
exit
end
configure terminal
router ospf 1
network 192.168.40.0 0.0.0.255 area 2
exit
end
show ip ospf neighbor
```

```
R6
configure terminal
router ospf 1
network 10.0.4.0 0.0.0.3 area 2
exit
end
configure terminal
router ospf 1
network 192.168.30.0 0.0.0.255 area 2
exit
end
show ip ospf neighbor
```

```
R6#
R6#
R6#
R6#
R6#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#router ospf 1
R6(config-router)#network 10.0.4.0 0.0.0.3 area 2
R6(config-router)#exit
R6(config)#end
*Mar 1 11:59:49.673: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.40.1 on FastEthernet2/0 from LOADING to FULL, Loading Done
R6(config)#end
R6#
*Mar 1 12:00:12.833: %SYS-5-CONFIG_I: Configured from console by console
R6#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.40.1	1	FULL/DR	00:00:35	10.0.4.2	FastEthernet2/0
10.0.10.2	1	FULL/DR	00:00:31	10.0.3.1	FastEthernet0/0

```
R6#
R6#
```

R7
configure terminal
router ospf 1
network 10.0.5.0 0.0.0.3 area 2
exit
end
show ip ospf neighbor

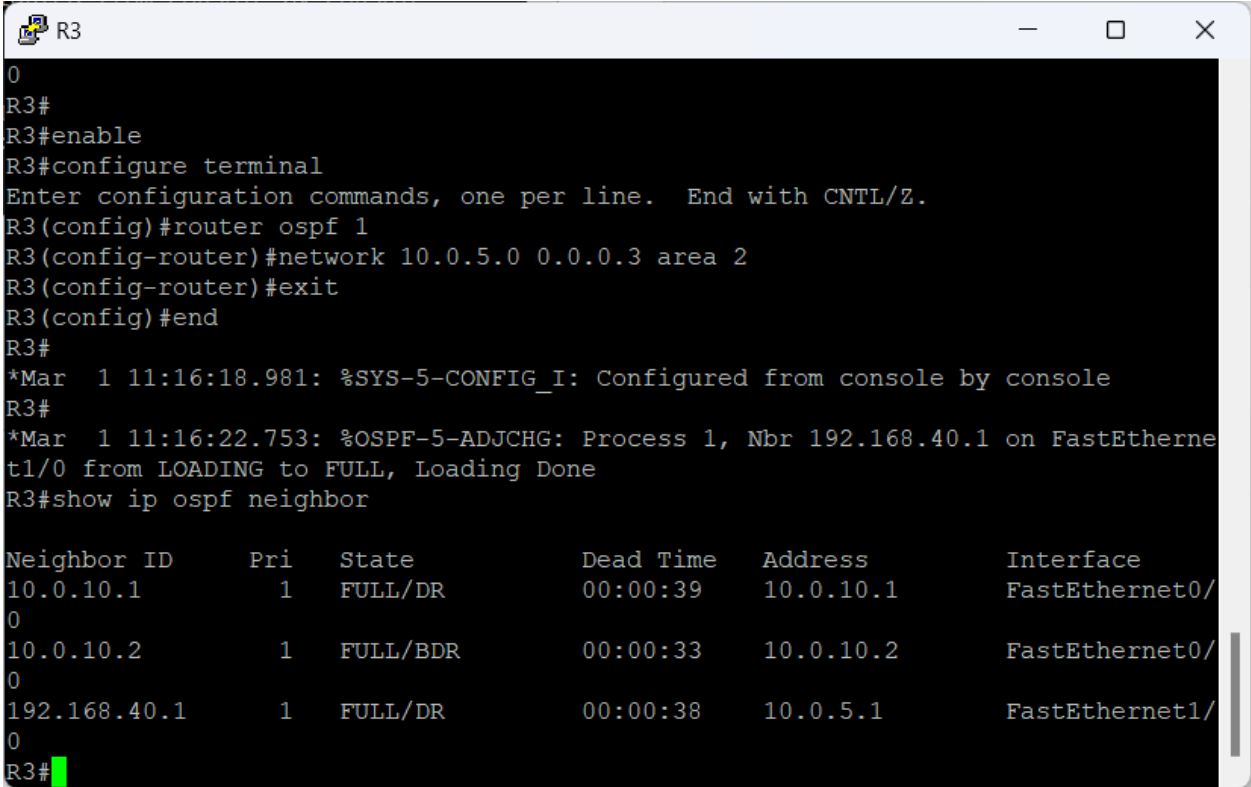
```
R7
R7(config)#router ospf 1
R7(config-router)#network 10.0.4.0 0.0.0.3 area 2
R7(config-router)#exit
R7(config)#end
R7#
*Mar 1 00:09:50.011: %SYS-5-CONFIG_I: Configured from console by console
R7#
*Mar 1 00:09:55.343: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.30.1 on FastEthernet2/0 from LOADING to FULL, Loading Done
R7#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#router ospf 1
R7(config-router)#network 10.0.5.0 0.0.0.3 area 2
R7(config-router)#exit
R7(config)#end
R7#
*Mar 1 00:10:05.859: %SYS-5-CONFIG_I: Configured from console by console
R7#
R7#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.30.1	1	FULL/DR	00:00:37	10.0.4.1	FastEthernet2/0

```
R7#
```

R3
enable
configure terminal
router ospf 1
network 10.0.5.0 0.0.0.3 area 2
exit
end

show ip ospf neighbor



```
0
R3#
R3#enable
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 10.0.5.0 0.0.0.3 area 2
R3(config-router)#exit
R3(config)#end
R3#
*Mar  1 11:16:18.981: %SYS-5-CONFIG_I: Configured from console by console
R3#
*Mar  1 11:16:22.753: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.40.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.0.10.1         1    FULL/DR         00:00:39    10.0.10.1      FastEthernet0/0
10.0.10.2         1    FULL/BDR        00:00:33    10.0.10.2      FastEthernet0/0
192.168.40.1      1    FULL/DR         00:00:38    10.0.5.1       FastEthernet1/0
R3#
```

R3
enable
configure terminal
router ospf 1
network 10.0.6.0 0.0.0.3 area 1
exit
end

show ip ospf neighbor

R8
enable
configure terminal
router ospf 1
network 10.0.6.0 0.0.0.3 area 1
exit
end
configure terminal
router ospf 1
network 192.168.50.0 0.0.0.255 area 1

```
exit
end
show ip ospf neighbor
```

Делаем Area 1 полностью тупиковой

На R3

```
enable
configure terminal
router ospf 1
area 1 stub no-summary
exit
end
```

На R8

```
enable
configure terminal
router ospf 1
area 1 stub
exit
end
```

После этого на обоих маршрутизаторах

```
show ip ospf
```

Редистрибуция между RIP v2 и OSPF

Сохраняем текущую рабочую конфигурацию R1

```
copy running-config startup-config
```

На R1

```
configure terminal
router ospf 1
redistribute rip subnets
exit
end
```

На R2

```
show ip route
```

```
R2
t1/0 from LOADING to FULL, Loading Done
R2#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O E2 192.168.10.0/24 [110/20] via 10.0.10.1, 00:01:51, FastEthernet0/0
O E2 192.168.20.0/24 [110/20] via 10.0.10.1, 00:01:51, FastEthernet0/0
    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
C     10.0.10.0/24 is directly connected, FastEthernet0/0
O E2   10.0.2.0/30 [110/20] via 10.0.10.1, 00:01:51, FastEthernet0/0
C     10.0.3.0/30 is directly connected, FastEthernet1/0
O E2   10.0.1.0/30 [110/20] via 10.0.10.1, 00:01:51, FastEthernet0/0
O IA   10.0.6.0/30 [110/2] via 10.0.10.3, 00:02:02, FastEthernet0/0
O     10.0.4.0/30 [110/2] via 10.0.3.2, 00:38:32, FastEthernet1/0
O     10.0.5.0/30 [110/3] via 10.0.3.2, 00:38:32, FastEthernet1/0
R2#
R2#
```

Теперь вторая сторона: OSPF - RIP

Делать нужно на R1, потому что именно R1 находится на границе RIP и OSPF.

На R1 вводим

configure terminal

router rip

redistribute ospf 1 metric 2

exit

end

show ip protocols

```
R1
R1(config-router)#exit
R1(config)#end
R1#
*Mar  1 14:23:15.328: %SYS-5-CONFIG_I: Configured from console by console
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router rip
R1(config-router)#redistribute ospf 1 metric 2
R1(config-router)#exit
R1(config)#end
R1#
*Mar  1 14:31:49.696: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 1 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip, ospf 1
  Default version control: send version 2, receive version 2
    Interface          Send Recv Triggered RIP Key-chain
  FastEthernet0/0      2      2
  FastEthernet1/0      2      2
Automatic network summarization is not in effect
```

Теперь на R4, R5

show ip route rip

```
PuTTY (inactive)
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, FastEthernet0/0
R    192.168.20.0/24 [120/1] via 10.0.1.2, 00:00:28, FastEthernet1/0
    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
R    10.0.10.0/24 [120/2] via 10.0.1.2, 00:00:28, FastEthernet1/0
R    10.0.2.0/30 [120/1] via 10.0.1.2, 00:00:28, FastEthernet1/0
C    10.0.1.0/30 is directly connected, FastEthernet1/0
R4#
R4#show ip route rip
R    192.168.20.0/24 [120/1] via 10.0.1.2, 00:00:25, FastEthernet1/0
    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
R    10.0.10.0/24 [120/2] via 10.0.1.2, 00:00:25, FastEthernet1/0
R    10.0.2.0/30 [120/1] via 10.0.1.2, 00:00:25, FastEthernet1/0
R    10.0.3.0/30 [120/3] via 10.0.1.2, 00:00:25, FastEthernet1/0
R    10.0.6.0/30 [120/3] via 10.0.1.2, 00:00:25, FastEthernet1/0
R    10.0.4.0/30 [120/3] via 10.0.1.2, 00:00:25, FastEthernet1/0
R    10.0.5.0/30 [120/3] via 10.0.1.2, 00:00:25, FastEthernet1/0
R4#
R4#
```

```
R5
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

R    192.168.10.0/24 [120/1] via 10.0.1.1, 00:00:25, FastEthernet1/0
C    192.168.20.0/24 is directly connected, FastEthernet0/0
     10.0.0.0/30 is subnetted, 2 subnets
C      10.0.2.0 is directly connected, FastEthernet2/0
C      10.0.1.0 is directly connected, FastEthernet1/0
R5#show ip route rip
R    192.168.10.0/24 [120/1] via 10.0.1.1, 00:00:11, FastEthernet1/0
     10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
R      10.0.10.0/24 [120/1] via 10.0.2.2, 00:00:23, FastEthernet2/0
R      10.0.3.0/30 [120/2] via 10.0.2.2, 00:00:23, FastEthernet2/0
R      10.0.6.0/30 [120/2] via 10.0.2.2, 00:00:23, FastEthernet2/0
R      10.0.4.0/30 [120/2] via 10.0.2.2, 00:00:23, FastEthernet2/0
R      10.0.5.0/30 [120/2] via 10.0.2.2, 00:00:23, FastEthernet2/0
R5#
```

Ping VPC «все между всеми» в обе стороны

На PC1

ping 192.168.20.2
ping 192.168.30.2
ping 192.168.40.2
ping 192.168.50.2

```
PC1 - PuTTY
PC1> ping 192.168.20.2
64 bytes from 192.168.20.2 icmp_seq=1 ttl=62 time=39.629 ms
64 bytes from 192.168.20.2 icmp_seq=2 ttl=62 time=33.629 ms
64 bytes from 192.168.20.2 icmp_seq=3 ttl=62 time=36.123 ms
64 bytes from 192.168.20.2 icmp_seq=4 ttl=62 time=22.675 ms
64 bytes from 192.168.20.2 icmp_seq=5 ttl=62 time=36.656 ms

PC1> ping 192.168.30.2
64 bytes from 192.168.30.2 icmp_seq=1 ttl=59 time=74.812 ms
64 bytes from 192.168.30.2 icmp_seq=2 ttl=59 time=71.196 ms
64 bytes from 192.168.30.2 icmp_seq=3 ttl=59 time=82.572 ms
64 bytes from 192.168.30.2 icmp_seq=4 ttl=59 time=83.997 ms
64 bytes from 192.168.30.2 icmp_seq=5 ttl=59 time=70.335 ms

PC1> ping 192.168.40.2
64 bytes from 192.168.40.2 icmp_seq=1 ttl=59 time=73.657 ms
64 bytes from 192.168.40.2 icmp_seq=2 ttl=59 time=83.810 ms
64 bytes from 192.168.40.2 icmp_seq=3 ttl=59 time=73.391 ms
64 bytes from 192.168.40.2 icmp_seq=4 ttl=59 time=62.522 ms
64 bytes from 192.168.40.2 icmp_seq=5 ttl=59 time=74.387 ms

PC1> ping 192.168.50.2
64 bytes from 192.168.50.2 icmp_seq=1 ttl=59 time=76.861 ms
64 bytes from 192.168.50.2 icmp_seq=2 ttl=59 time=90.029 ms
64 bytes from 192.168.50.2 icmp_seq=3 ttl=59 time=61.337 ms
64 bytes from 192.168.50.2 icmp_seq=4 ttl=59 time=63.219 ms
64 bytes from 192.168.50.2 icmp_seq=5 ttl=59 time=73.532 ms

PC1>
```


Ha PC2

ping 192.168.10.2

ping 192.168.30.2

ping 192.168.40.2

ping 192.168.50.2

```
PC2 - PuTTY
PC2> ping 192.168.10.2
84 bytes from 192.168.10.2 icmp_seq=1 ttl=62 time=31.613 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=62 time=22.538 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=62 time=28.763 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=62 time=22.487 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=62 time=24.925 ms

PC2> ping 192.168.30.2
84 bytes from 192.168.30.2 icmp_seq=1 ttl=60 time=63.174 ms
84 bytes from 192.168.30.2 icmp_seq=2 ttl=60 time=62.131 ms
84 bytes from 192.168.30.2 icmp_seq=3 ttl=60 time=61.515 ms
84 bytes from 192.168.30.2 icmp_seq=4 ttl=60 time=51.541 ms
84 bytes from 192.168.30.2 icmp_seq=5 ttl=60 time=53.913 ms

PC2> ping 192.168.40.2
84 bytes from 192.168.40.2 icmp_seq=1 ttl=60 time=43.374 ms
84 bytes from 192.168.40.2 icmp_seq=2 ttl=60 time=43.959 ms
84 bytes from 192.168.40.2 icmp_seq=3 ttl=60 time=52.796 ms
84 bytes from 192.168.40.2 icmp_seq=4 ttl=60 time=53.975 ms
84 bytes from 192.168.40.2 icmp_seq=5 ttl=60 time=60.266 ms

PC2> ping 192.168.50.2
84 bytes from 192.168.50.2 icmp_seq=1 ttl=60 time=60.869 ms
84 bytes from 192.168.50.2 icmp_seq=2 ttl=60 time=46.934 ms
84 bytes from 192.168.50.2 icmp_seq=3 ttl=60 time=55.154 ms
84 bytes from 192.168.50.2 icmp_seq=4 ttl=60 time=61.828 ms
84 bytes from 192.168.50.2 icmp_seq=5 ttl=60 time=53.622 ms

PC2> █
```

Ha PC3

ping 192.168.10.2

ping 192.168.20.2

ping 192.168.40.2

ping 192.168.50.2

```
PC3 - PuTTY
PC3> ping 192.168.10.2
84 bytes from 192.168.10.2 icmp_seq=1 ttl=59 time=91.634 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=59 time=84.870 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=59 time=73.407 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=59 time=65.524 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=59 time=73.899 ms

PC3> ping 192.168.20.2
84 bytes from 192.168.20.2 icmp_seq=1 ttl=60 time=52.554 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=60 time=51.964 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=60 time=51.318 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=60 time=52.768 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=60 time=54.150 ms

PC3> ping 192.168.40.2
84 bytes from 192.168.40.2 icmp_seq=1 ttl=62 time=39.224 ms
84 bytes from 192.168.40.2 icmp_seq=2 ttl=62 time=30.685 ms
84 bytes from 192.168.40.2 icmp_seq=3 ttl=62 time=21.779 ms
84 bytes from 192.168.40.2 icmp_seq=4 ttl=62 time=32.309 ms
84 bytes from 192.168.40.2 icmp_seq=5 ttl=62 time=29.439 ms

PC3> ping 192.168.50.2
84 bytes from 192.168.50.2 icmp_seq=1 ttl=60 time=60.476 ms
84 bytes from 192.168.50.2 icmp_seq=2 ttl=60 time=51.608 ms
84 bytes from 192.168.50.2 icmp_seq=3 ttl=60 time=56.096 ms
84 bytes from 192.168.50.2 icmp_seq=4 ttl=60 time=50.198 ms
84 bytes from 192.168.50.2 icmp_seq=5 ttl=60 time=42.282 ms

PC3> █
```

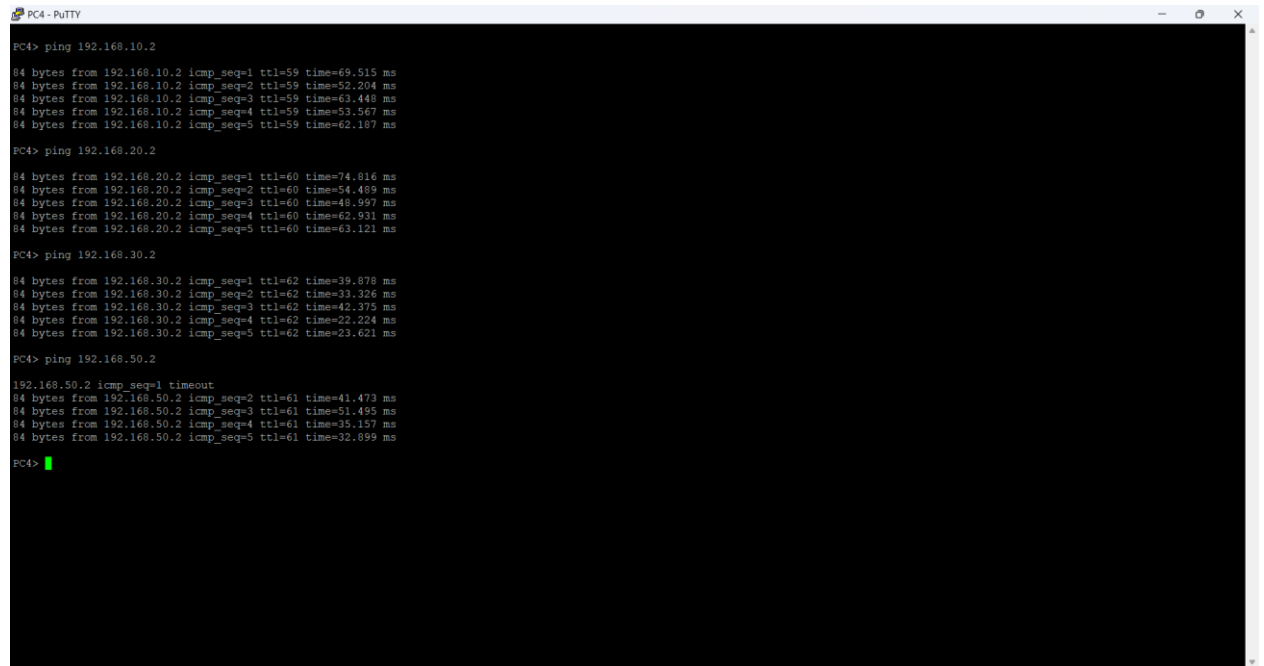
Ha PC4

ping 192.168.10.2

ping 192.168.20.2

ping 192.168.30.2

ping 192.168.50.2



```
PC4 - PuTTY
PC4> ping 192.168.10.2
64 bytes from 192.168.10.2 icmp_seq=1 ttl=59 time=69.515 ms
64 bytes from 192.168.10.2 icmp_seq=2 ttl=59 time=52.204 ms
64 bytes from 192.168.10.2 icmp_seq=3 ttl=59 time=63.448 ms
64 bytes from 192.168.10.2 icmp_seq=4 ttl=59 time=53.567 ms
64 bytes from 192.168.10.2 icmp_seq=5 ttl=59 time=62.187 ms

PC4> ping 192.168.20.2
64 bytes from 192.168.20.2 icmp_seq=1 ttl=60 time=74.816 ms
64 bytes from 192.168.20.2 icmp_seq=2 ttl=60 time=54.489 ms
64 bytes from 192.168.20.2 icmp_seq=3 ttl=60 time=48.997 ms
64 bytes from 192.168.20.2 icmp_seq=4 ttl=60 time=62.931 ms
64 bytes from 192.168.20.2 icmp_seq=5 ttl=60 time=63.121 ms

PC4> ping 192.168.30.2
64 bytes from 192.168.30.2 icmp_seq=1 ttl=62 time=39.878 ms
64 bytes from 192.168.30.2 icmp_seq=2 ttl=62 time=33.326 ms
64 bytes from 192.168.30.2 icmp_seq=3 ttl=62 time=42.375 ms
64 bytes from 192.168.30.2 icmp_seq=4 ttl=62 time=22.224 ms
64 bytes from 192.168.30.2 icmp_seq=5 ttl=62 time=23.621 ms

PC4> ping 192.168.50.2
192.168.50.2 icmp_seq=1 timeout
64 bytes from 192.168.50.2 icmp_seq=2 ttl=61 time=41.473 ms
64 bytes from 192.168.50.2 icmp_seq=3 ttl=61 time=51.495 ms
64 bytes from 192.168.50.2 icmp_seq=4 ttl=61 time=35.157 ms
64 bytes from 192.168.50.2 icmp_seq=5 ttl=61 time=32.899 ms

PC4>
```

Ha PC5

ping 192.168.10.2

ping 192.168.20.2

ping 192.168.30.2

ping 192.168.40.2

```
PCS - PuTTY

PC5> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=59 time=90.960 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=59 time=71.529 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=59 time=63.883 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=59 time=72.326 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=59 time=66.344 ms

PC5> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=60 time=55.707 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=60 time=60.867 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=60 time=54.319 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=60 time=52.703 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=60 time=51.658 ms

PC5> ping 192.168.30.2

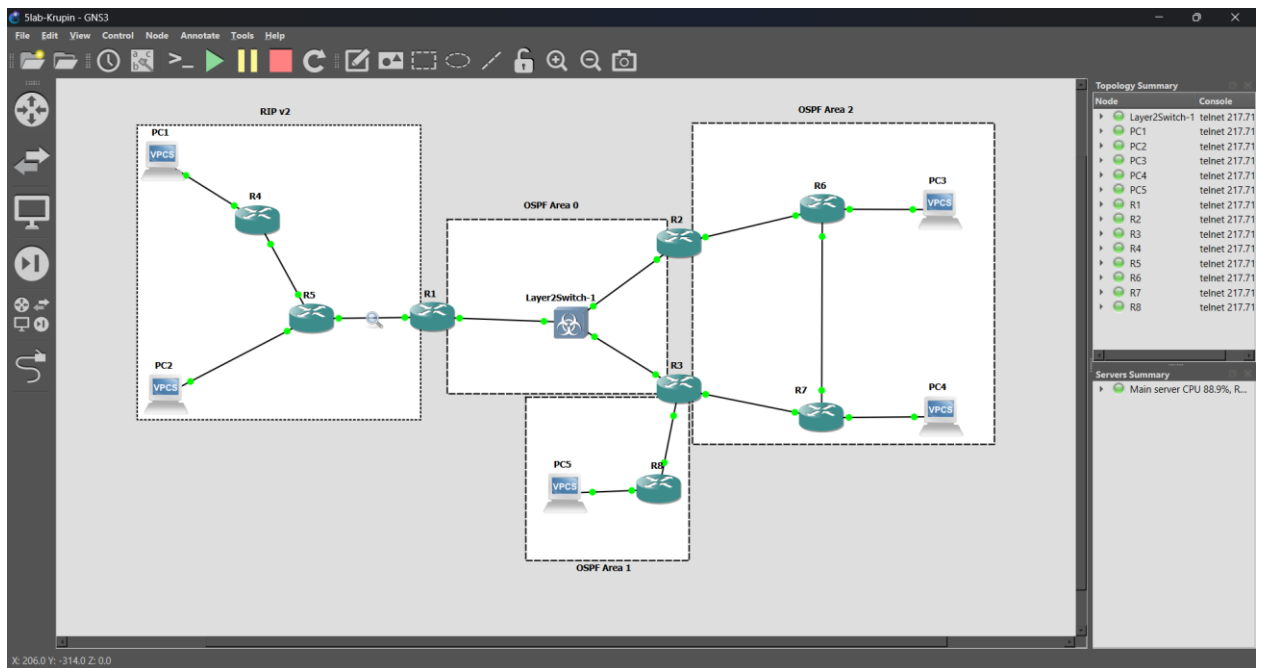
84 bytes from 192.168.30.2 icmp_seq=1 ttl=60 time=65.422 ms
84 bytes from 192.168.30.2 icmp_seq=2 ttl=60 time=59.209 ms
84 bytes from 192.168.30.2 icmp_seq=3 ttl=60 time=44.214 ms
84 bytes from 192.168.30.2 icmp_seq=4 ttl=60 time=54.634 ms
84 bytes from 192.168.30.2 icmp_seq=5 ttl=60 time=75.045 ms

PC5> ping 192.168.40.2

84 bytes from 192.168.40.2 icmp_seq=1 ttl=61 time=46.483 ms
84 bytes from 192.168.40.2 icmp_seq=2 ttl=61 time=32.092 ms
84 bytes from 192.168.40.2 icmp_seq=3 ttl=61 time=36.589 ms
84 bytes from 192.168.40.2 icmp_seq=4 ttl=61 time=32.118 ms
84 bytes from 192.168.40.2 icmp_seq=5 ttl=61 time=32.521 ms

PC5>
```

Перехват сообщений RIP v2 и OSPF в Wireshark



*Standard input (R5 FastEthernet2/0 to R1 FastEthernet0/0)

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

rip

No.	Time	Delta	Source	Destination	Protocol	Length	Info
2	3.335495		10.0.2.2	224.0.0.9	RIPv2	206	Response
7	15.451242		12.115747 10.0.2.1	224.0.0.9	RIPv2	106	Response
12	30.378890		14.927648 10.0.2.2	224.0.0.9	RIPv2	206	Response
15	44.585365		14.206475 10.0.2.1	224.0.0.9	RIPv2	106	Response
18	57.401691		12.816326 10.0.2.2	224.0.0.9	RIPv2	206	Response
24	75.884382		18.482691 10.0.2.1	224.0.0.9	RIPv2	106	Response
27	86.640991		10.756609 10.0.2.2	224.0.0.9	RIPv2	206	Response

Frame 2: Packet, 206 bytes on wire (1648 bits), 206 bytes captured (1648 bits) on interface -, id 0

Ethernet II, Src: ca:01:03:c:f0:00 (ca:01:03:c:f0:00), Dst: IPv4mcast_09 (01:00:5e:00:00:09)

Internet Protocol Version 4, Src: 10.0.2.2, Dst: 224.0.0.9

User Datagram Protocol, Src Port: 520, Dst Port: 520

Routing Information Protocol

Command: Response (2)

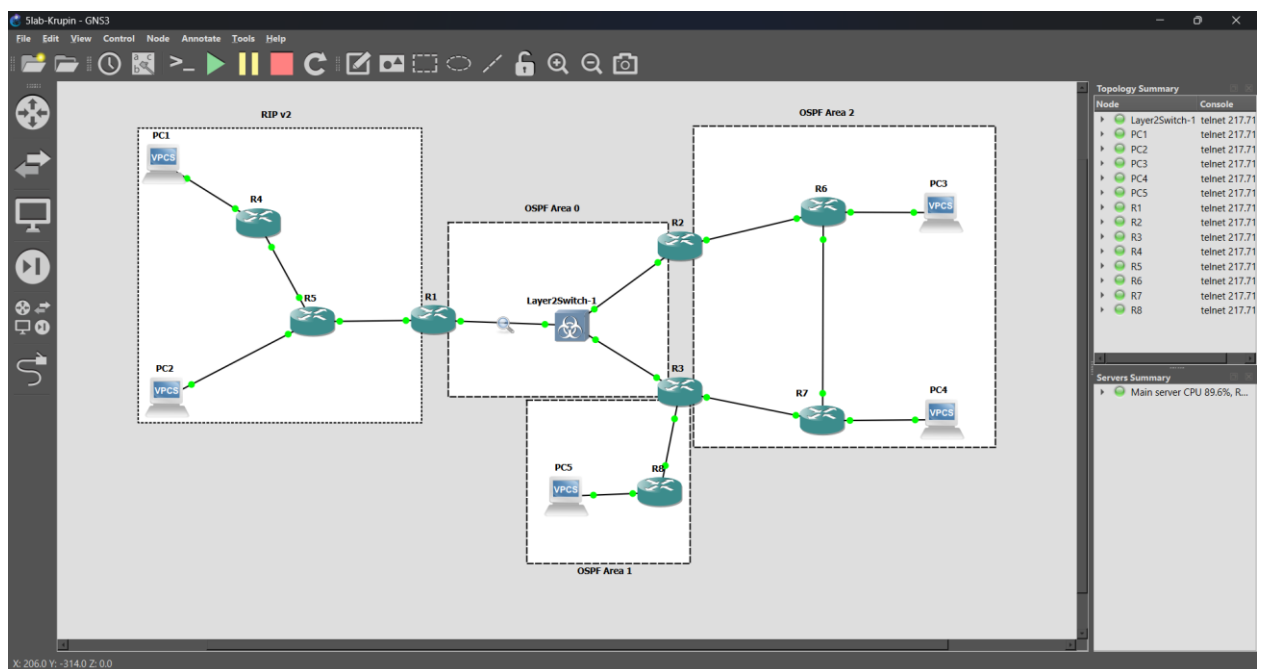
Version: RIPv2 (2)

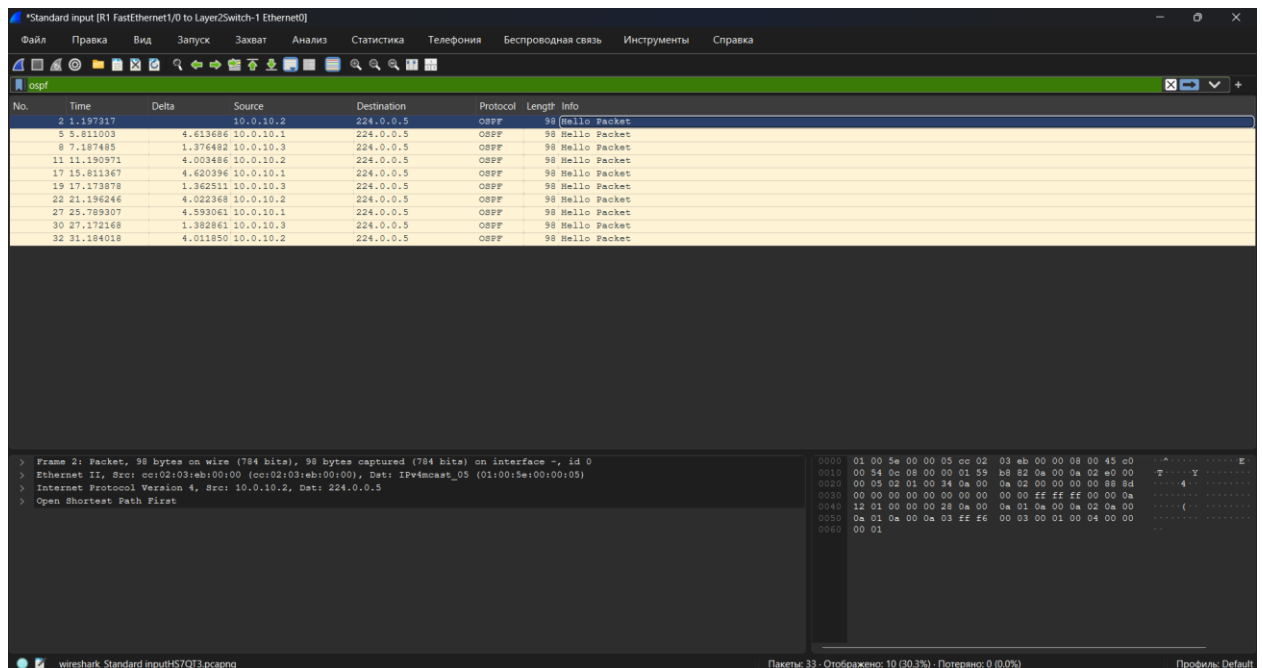
- IP Address: 10.0.3.0, Metric: 2
- IP Address: 10.0.4.0, Metric: 2
- IP Address: 10.0.5.0, Metric: 2
- IP Address: 10.0.6.0, Metric: 2
- IP Address: 10.0.10.0, Metric: 1
- IP Address: 192.168.30.0, Metric: 2
- IP Address: 192.168.40.0, Metric: 2
- IP Address: 192.168.50.0, Metric: 2

Text Item (text), 20 байт

Пакеты: 27 - Отображено: 7 (25.9%) - Потери: 0 (0.0%) Профиль: Default

В перехваченном пакете определён протокол RIPv2. Сообщение содержит информацию о доступных IPv4-сетях, масках подсетей и метриках маршрутов.





В перехваченном пакете определён протокол OSPF версии 2. Тип сообщения — Hello. Hello-сообщения используются маршрутизаторами для обнаружения соседей и поддержания OSPF-соседства.

Сохранить таблицы маршрутизации всех маршрутизаторов

Нужно на каждом маршрутизаторе:

show ip route

Сохранить файлы конфигураций устройств

Нужно на каждом маршрутизаторе:

show running-config